$$(\Lambda\alpha^{k}.v)[\tau] \quad \rightarrow \quad v[\alpha := \tau] \qquad \text{Step-E-TApp}$$

$$\texttt{handler}^{\varepsilon(h)}\ v \quad \rightarrow \quad \texttt{handle}^{\varepsilon(h,(v\ ()))} \qquad \text{Step-E-Handler}$$

$$\texttt{handle}^{\varepsilon(h,v)} \quad \rightarrow \quad v \qquad \text{Step-E-Return}$$

$$\frac{\mathsf{op} \notin \texttt{bop}(E) \wedge (\mathsf{op} \rightarrow f) \in h \qquad \mathsf{op} : \forall\alpha.\tau_1 \rightarrow \tau_2 \in \Sigma(l) \qquad k = \mathsf{lambda}^{\varepsilon} x : \tau_2[\alpha := \tau].\texttt{handle}^{\varepsilon(h,E[x])}}{\texttt{handle}^{\varepsilon\left(h,E\left[\texttt{perform}^{\varepsilon}_{p(\mathsf{op},\tau:\tau)}\ v\right]\right)} \quad \rightarrow \quad f[\tau]\ v\ k} \text{ Step-E-Perform}$$

$$\frac{E \neq \square}{E\left[\texttt{error}_p\right], \sigma \Longrightarrow \texttt{error}_p, \sigma} \text{ Step-E-Error} \qquad \frac{e_1 \rightarrow e_2}{E[e_1], \sigma \Longrightarrow E[e_2], \sigma} \text{ Step-E}$$

Helper functions

$$\begin{array}{lcl}
\texttt{bop}(\square) & = & \varnothing \\
\texttt{bop}(E\ e) & = & \texttt{bop}(E) \\
\vdots & \vdots & \vdots \\
\texttt{bop}\left(\texttt{handle}^{\varepsilon(h,E)}\right) & = & \texttt{bop}(E) \cup \{\mathsf{op} \mid (\mathsf{op} \rightarrow f) \in h\}
\end{array}$$

Listing 1: Reduction Rules for `Effects`